Test Case	Input Scenario	Expected Result	Actual Result	Status
User Registration	Valid email and password	Account created successfully	Account created	Passed
User Registration	Duplicate email	Error message shown	Error displayed	Passed
User Login	Correct credentials	Login successful, token generated	Login successful	Passed
User Login	Incorrect password	Access denied	Error displayed	Passed
Submit Issue	Valid product data	Product created and stored	Product created	Passed
Get All Issues	Valid materials and percentages	Batch created successfully	Batch created	Passed
Update Issue Status	No change in materials	Auto verification	Verified automatically	Passed
Add Comment	Minor change in composition	AI scoring triggered	Score generated	Passed
Unauthorized Access	Major change	Lab verification required	Marked for lab	Passed

Role-Based Access	Valid batch data	Score and insights generated	Score displayed	Passed
Error Handling	Valid report data	Report submitted	Report created	Passed
Admin Approval	Approve lab report	Batch verified	Verified	Passed
Admin Rejection	Reject lab report	Batch rejected	Rejected	Passed